

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: B.Tech.

Branch: Common To All Branches

Semester: II

Subject Code & Name: BTES203 & Engineering Mechanics

Max Marks: 60

Date:04/07/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		CO	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Two forces 2N and 4N act at a point on a body. The resultant when they act 60° is	CO1	1
	a. $(10)^{1/2}$ b. $(28)^{1/2}$ c. $(6)^{1/2}$ d. $(8)^{1/2}$		
2	What are the X and Y components of point P for the force system shown below?	CO1	1
	a. X = 186.00 N, b. X = 464.23 N, c. X = 466.12 N, d. X = 110.15 N, Y = 464 N Y = 185 N Y = -180 N Y = - 320 N		
3	If the resultant of two equal forces has the same magnitude as either of the forces, then the angle between the two forces is	CO1	1
	a. 30° b. 60° c. 90° d. 120°		
4	The reaction at the roller supported end of a beam is always	CO2	1
	a. vertical b. horizontal c. inclined d. zero		
5	Angle of friction is given as	CO3	1
	a. $\sin^{-1}(\mu)$ b. $\cos^{-1}(\mu)$ c. $\tan^{-1}(\mu)$ d. $\sin(\mu)$		
6	A simply supported beam of span L is loaded with uniformly varying load varies from zero at left support to maximum intensity (w) at right support, the support reactions are	CO3	1
	a. $WL/3$ b. $WL/6$ c. Both are correct d. Both are wrong		
7	When a car moves at constant speed around a curved path its velocity is	CO4	1
	a. zero b. constant c. change in magnitude d. changes in direction		
8	Area under velocity time curve over a given interval of time represents	CO4	1
	a. Acceleration b. Momentum c. Velocity d. Displacement		
9	When a body is lifted up, work done by force of gravity is	CO4	1
	a. Positive b. Negative c. Zero d. None		

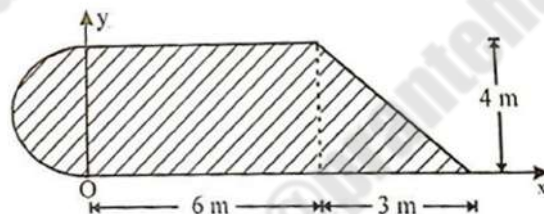
10	A missile is fired so as to reach maximum range then maximum height reached is	CO4	1
	a. Same as range b. 3/4th range c. Half of range d. 1/4th of range		
11	When speed of particle is doubled, its kinetic energy	CO5	1
	a. Remains same b. Increase doubled c. Increase three times d. Increase four times		
12	Impulse momentum principal says	CO5	1
	a. Change in K.E.=Work done b. Impulse of force system =Change in momentum c. Change in impulsive force=Change in K.E. d. Change in P.E. =Change in K.E.		

Q. 2 Solve the following. **12**

- | | | |
|---|-----|---|
| A) Explain in detail classification of force system. | CO1 | 6 |
| B) State and explain 1) Lami's Theorem 2) Varignon's Theorem. | CO1 | 6 |

Q.3 Solve the following. **12**

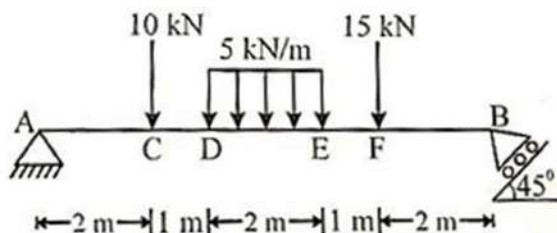
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|---|-----|---|
| A) Locate the centroid of the shaded area shown in the figure | CO2 | 6 |
|---|-----|---|



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| B) State and explain analytical and graphical conditions of equilibrium. | CO2 | 6 |
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Q. 4 Solve Any Two of the following. **12**

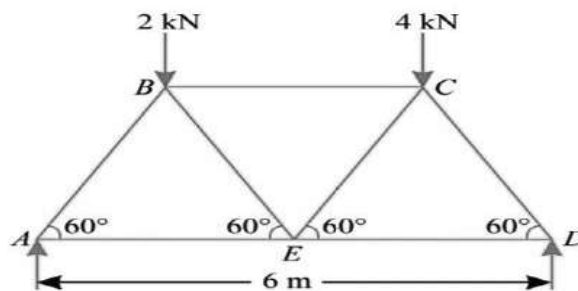
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|---|-----|---|
| A) Find support reactions for the beam loaded as shown in the figure. | CO3 | 6 |
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| B) State and explain different types of trusses. | CO3 | 6 |
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- C) Fig. shows a girder consisting of seven members each of 3 m length freely supported at its end points. The girder is loaded at B and C as shown. Find the forces in all the members of the girder indicating their nature.

CO3 6

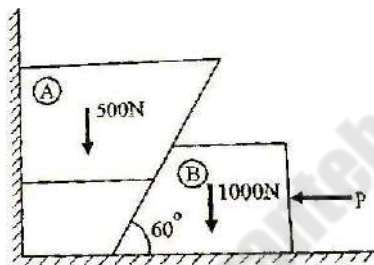


Q.5 Solve Any Two of the following.

12

- A) Two blocks A and B are resting against a wall and the floor as shown in fig. Find horizontal force P applied to the lower block that will hold the system in equilibrium. Take $\mu=0.25$ at floor, $\mu=0.3$ at wall and $\mu=0.2$ between the blocks

CO4 6



- B) The velocity of a particle moving along x axis is defined by $v=kx^3-4x^2+6x$ where 'v' is in m/s, 'x' is in meters and 'k' is constant. If $k=1$, determine the value of acceleration when $x=2$ m.
- C) A body is projected at an angle such that its horizontal range is five times the maximum height attained. Determine its angle of projection.

CO4 6

CO4 6

Q. 6 Solve Any Two of the following.

12

- A) 1) Explain D'Alembert's Principle
2) Define work and state and explain law of conservation of energy.
- B) An object weighing 5 N falls through a height of 40 m and buries itself 1.5 m deep in sand. Find the average resistance of penetration using work energy principle.
- C) A particle moves along the path $x=2t$, $y=2t^2$ where t is the time seconds and x, y are in meters. What work is done in the interval from $t=0$ to $t=3$ seconds by a force whose components are $F_x=(2+t)$ and $F_y=2t^2$? Forces are in Newtons

CO5 6

CO5 6

CO5 6

*** End ***